



Day 17: Encoder-Decoder Models (BART, T5, FLAN-T5)



1. Quick Revision (Very Important Start)

Before jumping in, always connect with previous days:

Type	Model	Work
Encoder	BERT	Understand text
Decoder	GPT	Generate text
Encoder-Decoder	T5, BART	Understand + Generate



- Encoder → reads entire input
- Decoder → generates output step-by-step
- Encoder-Decoder → does BOTH 🔥



2. What is Encoder-Decoder Architecture?



It is a sequence-to-sequence (Seq2Seq) model.

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◆ Flow:

Input → Encoder → Context Vector → Decoder → Output



Simple Example (Translation)

Input:

“Hello, how are you?”

Output:

“नमस्ते, आप कैसे हैं?”

◆ How it works:



Encoder:

- Reads full sentence
- Understands meaning
- Converts into context representation



Decoder:

- Takes context
- Generates output word-by-word



Core Idea:



“First understand → then generate”

3. Important Concept: Attention (Bridge)

Without attention:

- Decoder forgets long sentences ❌

With attention:

- Decoder focuses on relevant words ✅

👉 Example:

While translating “apple”, it focuses only on that word.

4. BART (Bidirectional + Auto-Regressive Transformer)

◆ Idea:

👉 Combine:

- BERT (Encoder power)
- GPT (Decoder power)

◆ How BART is trained:

- Input is **corrupted text**
- Model learns to **reconstruct original text**

◆ Example:

Input:

“The cat ____ on the mat”

Output:

“The cat sat on the mat”

◆ Use Cases:

- Text summarization
- Text correction
- Denoising

◆ Key Feature:

👉 Strong understanding + generation

🔥 5. T5 (Google)

◆ Most IMPORTANT model for interviews 🔥

◆ Core Idea:

👉 "Everything is TEXT → TEXT"

◆ Example Tasks:

Task	Input	Output
Translation	translate English to German: Hello	Hallo
Summarization	summarize: long text	short summary
QA	question: ... context: ...	answer

◆ Why Powerful?

- One model → all NLP tasks
- No architecture change needed

◆ Key Advantage:

👉 Standardized format (text-to-text)

Step 1: What T5 can do (but struggles with)

T5 is powerful because it converts everything into:

👉 **Text** → **Text**

But here's the real issue 👉

❌ Problem in T5 (Core Issue)

👉 T5 is **not naturally good at understanding human-style instructions**

It works best when:

- Input is in a **very specific format**
 - Task is **clearly structured like training data**
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🔴 Example 1 (Real Confusion Case)

Input:

👉 "Explain AI in simple words"

T5 Behavior:

- Might give:
- incomplete answer
- generic sentence
- sometimes irrelevant

👉 Because:

- It's thinking: *"What task is this? summarization? QA? definition?"*
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🔴 Example 2 (Format Dependency)

Input:

👉 "Hello in German"

T5 Output:

👉 ❌ May fail or give wrong output

But if you change format:

👉 "translate English to German: Hello"

T5 Output:

👉 ✅ "Hallo"

🔥 Root Problem (Very Important)

👉 T5 needs **task-specific prompts / formats**

It does NOT truly understand:

“What does the user want?”

🚀 Step 2: What FLAN-T5 Changed

FLAN-T5 fixed this by:

👉 Training the model on **thousands of instructions written like humans speak**

⚙️ What Exactly Was Done?

Instead of training like this:

👉 Input: “translate English to German: Hello”

FLAN training includes:



- “Translate this sentence to German: Hello”
 - “Can you convert this to German? Hello”
 - “What is Hello in German?”
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🎯 Result:

👉 Model learns:

“Different ways humans ask the same thing”

